

CBSE Class 10 Mathematics — Introduction to Trigonometry

Time Allowed: 3 Hours

Maximum Marks: 80

General Instructions:

1. This question paper contains *questions. All questions are compulsory.*
1. Questions are grouped into sections A through E.
2. Use of calculators is NOT permitted.

Section A: Multiple Choice & Assertion-Reason (1 mark each)

1. Assertion (A): If $\sin \theta = \frac{1}{2}$, then $\theta = 30^\circ$. Reason (R): The value of $\sin \theta$ increases as θ increases from 0° to 90° .

(A) (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A)
(B) (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A)
(C) (C) Assertion (A) is true but Reason (R) is false
(D) (D) Assertion (A) is false but Reason (R) is true
2. Assertion (A): The value of $\cos \theta$ can never be 1.5. Reason (R): The range of $\cos \theta$ is $[-1, 1]$.

(A) (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A)
(B) (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A)
(C) (C) Assertion (A) is true but Reason (R) is false
(D) (D) Assertion (A) is false but Reason (R) is true
3. Assertion (A): $\sin^2 \theta + \cos^2 \theta = 1$ for all θ . Reason (R): $\tan^2 \theta - \sec^2 \theta = 1$.

(A) (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A)
(B) (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A)
(C) (C) Assertion (A) is true but Reason (R) is false
(D) (D) Assertion (A) is false but Reason (R) is true
4. Assertion (A): If $\tan A = \cot A$, then $A = 45^\circ$. Reason (R): $\tan A = \cot(90^\circ - A)$.

- (A) (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A)
- (B) (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A)
- (C) (C) Assertion (A) is true but Reason (R) is false
- (D) (D) Assertion (A) is false but Reason (R) is true
5. Assertion (A): The value of $\sin 0^\circ$ is 0. Reason (R): The value of $\cos 0^\circ$ is 0.
- (A) (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A)
- (B) (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A)
- (C) (C) Assertion (A) is true but Reason (R) is false
- (D) (D) Assertion (A) is false but Reason (R) is true

Section C: Short Answer I (3 marks each)

- If $\tan A = \frac{4}{3}$, find the value of $\sin A + \cos A$.
- Evaluate: $\frac{2 \tan^2 45^\circ + \cos^2 30^\circ - \sin^2 60^\circ}{1}$
- If $\sin(A - B) = \frac{1}{2}$ and $\cos(A + B) = \frac{1}{2}$, where $0^\circ < A + B \leq 90^\circ$ and $A > B$, find A and B .
- Express $\cos A$ in terms of $\tan A$.
- Evaluate $\frac{\tan 65^\circ}{\cot 25^\circ}$.

Section D: Long Answer (4-5 marks each)

- If $\tan(A + B) = \sqrt{3}$ and $\tan(A - B) = \frac{1}{\sqrt{3}}$, where $0^\circ < A + B \leq 90^\circ$ and $A > B$, find the values of A and B .
- If $3 \cot \theta = 2$, evaluate the value of $\frac{4 \sin \theta - 3 \cos \theta}{2 \sin \theta + 6 \cos \theta}$.
- Prove that $(\sin A + \csc A)^2 + (\cos A + \sec A)^2 = 7 + \tan^2 A + \cot^2 A$.
- If $\sin \theta + \sin^2 \theta = 1$, prove that $\cos^2 \theta + \cos^4 \theta = 1$.
- Evaluate: $\frac{\cos 45^\circ}{\sec 30^\circ + \csc 30^\circ}$.
- If $\sin \theta = \frac{3}{5}$, then the value of $\cos \theta$ is:
- The value of $\tan 45^\circ + \cot 45^\circ$ is:
- If $3 \tan \theta = 3$, then θ is equal to:

9. The value of $\frac{\sin 18^\circ}{\cos 72^\circ}$ is:
10. Which of the following is equal to $\sec^2 \theta - 1$?
11. Prove that $\frac{\cos A - \sin A + 1}{\cos A + \sin A - 1} = \csc A + \cot A$, using the identity $\csc^2 A = 1 + \cot^2 A$.
12. If $\sin \theta + \cos \theta = \sqrt{3}$, then prove that $\tan \theta + \cot \theta = 1$.
13. Evaluate: $\frac{5 \cos^2 60^\circ + 4 \sec^2 30^\circ - \tan^2 45^\circ}{\sin^2 30^\circ + \cos^2 30^\circ}$
14. If $\sec \theta = x + \frac{1}{4x}$, prove that $\sec \theta + \tan \theta = 2x$ or $\frac{1}{2x}$.